XXIX ENMC ENCONTRO NACIONAL DE MODELAGEM COMPUTACIONAL
XVII ECTM ENCONTRO DE CIÊNCIA E TECNOLOGIA DE MATERIAIS

BIOSENSIA-DC: A DRUGCLIP-BASED COMPUTATIONAL PROTOTYPE FOR TARGET FISHING APPLIED TO BIORECEPTOR DISCOVERY

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Abstract.
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Keywords: *Biosensors, Target fishing, DrugCLIP, Contrastive learning, Chemoinformatics*

1. INTRODUCTION

1.1 Biosensors and the bioreceptor-selection problem

A biosensor can be understood as an analytical device that couples a biological recognition mechanism to a signal-transduction mechanism. The biological component is responsible for interacting with the analyte of interest, while the transducer converts that interaction into a measurable signal. In this work, we use the terms *bioreceptor* and *biorecognition element* in this general sense. Common examples include enzymes, antibodies, aptamers, receptors, and other biomolecular structures capable of selective molecular recognition (Eggins, 1996; Banica, 2012; Morales and Halpern, 2018). The nature of this recognition element strongly influences the sensitivity, selectivity, reproducibility, and practical applicability of the resulting biosensor.

Selecting a suitable bioreceptor is therefore one of the central design decisions in biosensor development. This decision is not merely a matter of choosing a molecule that can bind the analyte: the candidate must also be compatible with immobilization, signal generation, stability requirements, sample conditions, and the intended sensing platform. Enzyme-based biosensors, immunosensors, and aptamer-based sensors offer different advantages and limitations, but each case still requires the identification of a recognition mechanism that is selective enough for the target application (Marques and Yamanaka, 2008; Morales and Halpern, 2018).

The broader BioSensIA project addresses this problem as a computational discovery task. Given an analyte molecule, the goal is to identify proteins, protein domains, binding pockets, or other biomolecular structures that may interact with it and therefore deserve further investigation as candidate bioreceptors. In this formulation, the computational system is not expected to replace experimental validation. Rather, it is intended to reduce the search space:

instead of manually inspecting many unrelated biological databases and structural records, the user receives a prioritized list of candidates that can be examined with additional structural, biochemical, and bibliographic evidence. This ranking-oriented view is particularly useful when the biological interactions relevant to sensing are not already obvious from the literature.

BioSensIA-DC is one computational component of this broader workflow. It focuses on the structural target-fishing step: given a query molecule, it ranks candidate protein pockets according to a learned compatibility score. The prototype is based on DrugCLIP, a contrastive representation-learning model originally proposed for virtual screening, and on Uni-Mol-style three-dimensional molecular and pocket representations (Gao et al., 2023; Zhou et al., 2023). BioSensIA-DC uses DrugCLIP as a representation-learning foundation, but extends it with a new target-fishing workflow, data organization strategy, and retrieval implementation. In the original virtual-screening direction, a protein pocket is used as a query to rank molecules. BioSensIA-DC uses the same molecule–pocket compatibility idea in the reverse direction: the molecule becomes the query, and protein pockets become the candidates to be ranked. The resulting ranked list is intended to support bioreceptor discovery, not to provide a definitive proof of binding or biosensor performance.

1.2 Virtual screening, target fishing, and representation learning

Virtual screening and target fishing are closely related computational tasks, but they differ in the direction of the query. In structure-based virtual screening, the starting point is usually a biological target, often represented by a protein binding pocket, and the goal is to rank molecules from a candidate library according to their predicted compatibility with that target. In target fishing, the direction is reversed: the starting point is a molecule, and the goal is to identify proteins, protein domains, binding sites, or other biological targets that may interact with it. In both cases, the practical output is a ranked list, but the biological interpretation is different. Virtual screening asks which molecules may bind a given target; target fishing asks which targets may bind a given molecule.

Several *in silico* target-fishing strategies have been proposed, including ligand-based, receptor-based, pharmacophore-based, docking-based, and machine-learning approaches (Galati et al., 2021). PharmMapper is a representative example of reverse pharmacophore mapping. Given a query compound, it searches for good alignments between the compound and a database of pharmacophore models associated with protein targets (Liu et al., 2010). Its updated version expanded the target pharmacophore database and introduced statistically meaningful ranking scores for the identified targets (Wang et al., 2017). In simplified terms, PharmMapper asks whether the three-dimensional arrangement of chemical features in the query molecule can be fitted to pharmacophore patterns known or inferred for possible targets.

SwissTargetPrediction follows a different principle. It predicts probable targets of a small molecule by comparing the query compound to molecules with known bioactivity data. The method is based on the similarity principle: molecules that are similar in two-dimensional or three-dimensional chemical space are more likely to share biological targets (Daina et al., 2019). Thus, while PharmMapper is primarily organized around reverse matching against pharmacophore models, SwissTargetPrediction is organized around ligand similarity to compounds with experimentally known target annotations. Both tools are valuable references for BioSensIA because they address the same broad question: given a molecule, which biological targets should be considered plausible?

BioSensIA-DC differs from these tools in its representation-learning formulation. Instead of matching a query molecule against an explicit pharmacophore database, as in PharmMapper, or inferring targets from similarity to known ligands, as in SwissTargetPrediction, *BioSensIA-DC uses a learned compatibility model between molecules and protein pockets*. Its starting point is DrugCLIP, a contrastive learning model that represents molecules and protein pockets with separate encoders and trains them so that compatible molecule–pocket pairs are close in a shared latent space (Gao et al., 2023). DrugCLIP formulates virtual screening as a dense-retrieval problem: a pocket is encoded as a query vector, molecules are encoded as candidate vectors, and ranking is performed by comparing these vectors.

The encoders used by DrugCLIP are based on Uni-Mol, a three-dimensional molecular representation-learning framework that incorporates atom types and atomic coordinates (Zhou et al., 2023). A key feature of Uni-Mol is that it is designed to respect the geometry of molecular structures: its internal pair representations are built from interatomic distances, which are invariant to global translations and rotations of the molecule or pocket. At the same time, Uni-Mol includes an equivariant coordinate-prediction mechanism, so that when coordinates are predicted or updated, they transform consistently under rotations and translations of the input structure. In practical terms, this means that the representation is sensitive to the relative three-dimensional arrangement of atoms, but not to arbitrary choices of coordinate-system origin or orientation.

The central idea behind BioSensIA-DC is that the similarity score learned by DrugCLIP can be used in the reverse direction. If DrugCLIP can assign a compatibility score to a molecule–pocket pair for virtual screening, then the same score can be used to rank pockets for a fixed molecule. However, BioSensIA-DC is not merely a direct execution of the original DrugCLIP workflow. DrugCLIP provides the initial molecule–pocket representation framework, but the target-fishing formulation, candidate-pocket preparation, query construction, molecule-to-pocket retrieval pipeline, and initial diagnostic evaluation workflow were conceptualized and implemented from scratch as part of this work. The resulting prototype therefore adapts a virtual-screening representation model to the biosensor-oriented problem of prioritizing candidate bioreceptors for a query analyte.

1.3 Objectives and contributions

This paper describes BioSensIA-DC, a computational prototype for molecule-to-pocket target fishing in the context of bioreceptor discovery. The prototype extends the DrugCLIP representation framework beyond its original virtual-screening workflow. Instead of ranking molecules for a fixed protein pocket, BioSensIA-DC ranks candidate protein pockets for a fixed query molecule, producing a prioritized list of structural candidates for further investigation.

The implemented contributions include the construction of candidate-pocket files from DrugCLIP-compatible data, the construction of query-molecule files from SMILES strings, local DrugCLIP identifiers, PubChem CIDs, or compound names, and the implementation of molecule-to-pocket retrieval. The prototype also generates ranked pocket lists with compatibility scores, and provides an initial benchmarking and diagnostic workflow.

The present work should be read as a computational-prototype and methodology paper. Suitable benchmark validation has not yet been completed. Therefore, no claim is made that the current model is already a reliable target-fishing predictor. Instead, the paper reports the implemented workflow, the conceptual adaptation from virtual screening to target fishing, and the diagnostic issues that must be resolved before predictive performance can be assessed.

2. METHODOLOGY

2.1 Overview of the BioSensIA-DC workflow

BioSensIA-DC implements a molecule-to-pocket retrieval workflow for target fishing. The input is one or more query molecules, and the output is a ranked list of candidate protein pockets. The workflow reuses the molecule and pocket encoders from DrugCLIP, but reverses the practical direction of retrieval: instead of using a pocket to rank molecules, it uses a molecule to rank pockets.

The candidate protein pockets are stored in a library, which can be encoded once and cached for later searches. During retrieval, BioSensIA-DC encodes the query molecule, loads or computes the candidate-pocket embeddings, calculates molecule–pocket similarity scores, and sorts the candidate pockets by these scores. Figure 1 illustrates this workflow.

The resulting ranked list is saved for downstream analysis. Each entry contains a candidate pocket identifier and a compatibility score. This score is used only for ranking; it should not be interpreted as a calibrated binding affinity or as a direct physical measurement.

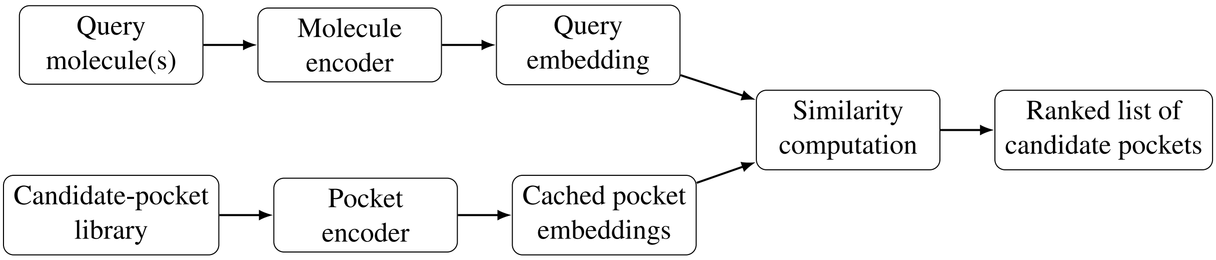

Figure 1: Overview of the BioSensIA-DC target fishing workflow. Query molecules are encoded and compared with cached candidate-pocket embeddings to produce a ranked list of protein pockets.

2.2 Base model: DrugCLIP and Uni-Mol

As Fig. 1 shows, DrugCLIP uses a *two-tower* architecture to represent molecule–pocket compatibility (Gao et al., 2023). One encoder maps each molecule to a latent vector, while a second encoder maps each protein pocket to another latent vector. The compatibility of a molecule–pocket pair is then computed by a similarity score between these two vectors, such as an inner product or cosine similarity.

Both encoders follow the Uni-Mol style of three-dimensional representation learning, in which the input is built from atom types and atomic coordinates rather than only from one-dimensional sequences or two-dimensional molecular graphs (Zhou et al., 2023). This allows the model to represent the local geometry of molecules and protein pockets, which is essential when compatibility depends on the three-dimensional arrangement of atoms.

Mathematically, we have the similarity score $s(p, m)$ defined as the scalar product

$$s(p, m) = g_\phi(x^p)^T \cdot f_\theta(x^m) \tag{1}$$

where p denotes a protein pocket, m denotes a molecule, and x^p and x^m denote their input representations. Furthermore, g_ϕ is the pocket encoder (parameterized by a set ϕ of parameters), and f_θ is the molecule encoder (parameterized by a set θ of parameters). In the current DrugCLIP implementation, both $g_\phi(x^p)$ and $f_\theta(x^m)$ are 128-dimensional vectors.

Figure 2 illustrates how DrugCLIP is trained: both encoders generate embeddings for *positive* pairs (x_i^p, x_i^m) and *negative* pairs $(x_i^p, x_j^m), i \neq j$, each pair containing a pocket and a molecule. Here, "positive" means having evidence of chemical affinity, and "negative" means lacking such evidence. Similarity scores are computed for each pair in the batch and a *loss function* is applied in order to modify the sets of trainable parameters ϕ and θ of the encoders.

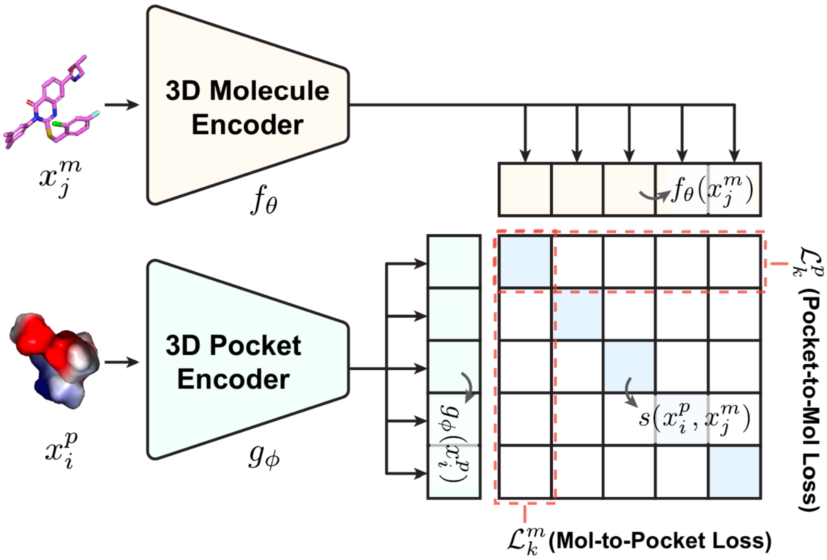

Figure 2: Overview of the DrugCLIP training process. Adapted from (Gao et al., 2023).

Formally, the DrugCLIP loss function has two terms: in a batch of N pairs, the *pocket-to-mol loss* for pocket x_k^p is defined as

$$\mathcal{L}_k^p(x_k^p, \{x_i^m\}_{i=1}^N) = -\log \frac{\exp(s(x_k^p, x_k^m)/\tau)}{\sum_i \exp(s(x_k^p, x_i^m)/\tau)} \tag{2}$$

and the *mol-to-pocket loss* for molecule x_k^m is defined as

$$\mathcal{L}_k^m = (x_k^m, \{x_i^p\}_{i=1}^N) = -\log \frac{\exp(s(x_k^p, x_k^m)/\tau)}{\sum_i \exp(s(x_i^p, x_k^m)/\tau)} \tag{3}$$

where $s(\cdot,\cdot)$ is defined in Eq.(1) and τ is a hyperparameter (called the *constrastive temperature*) that controls the smoothness of the softmax distribution. The final loss for a batch (which training seeks to minimize) is then

$$\mathcal{L} = \frac{1}{2}\sum_{k=1}^{N}(\mathcal{L}_k^p + \mathcal{L}_k^m) \tag{4}$$

The definitions above are important for the following section, where we describe the modifications we applied to the training procedure in order to fine-tune the model for the target-fishing problem.

2.3 Fine-tuning formulation for target fishing

Although virtual screening and target fishing use the same molecule–pocket compatibility score, they emphasize different retrieval directions. In virtual screening, a pocket is used as the query and molecules are ranked as candidates; in target fishing, a molecule is used as the query and pockets are ranked as candidates. This reversal is not only a change in input and output formatting. For the target-fishing use case, a query molecule may have several known or plausible protein pockets, corresponding to different proteins, homologs, isoforms, crystal structures, or binding-site variants. A fine-tuning objective that treats only one molecule–pocket pair as positive in each batch (as DrugCLIP does) may therefore penalize other known positives for the same molecule as implicit negatives. This observation motivates the target-fishing-oriented fine-tuning formulation described below.

Multiple positive pockets for a single molecule. In the training data for DrugCLIP, there were several molecules that bind more than one pocket, but this information was not taken into account. The idea is to use it in BioSensIA-DC.

We define the *set of known positive pockets* for a molecule m as

$$P^{+}(m) = \{p : (p, m) \in D\} \tag{5}$$

where D is the set of observed positive pocket-molecule pairs in the training data.

Molecule-centered sampling for batch building. During training of BioSensIA-DC, when we build a batch, we first sample a set of distinct molecules $\{m_1, m_2, \cdots, m_N\}$ and then, for each molecule m_i in the set, we sample (preferably two or more) positive pockets from $P^{+}(m_i)$. To keep track of the positive pairs in the batch, we also define a *positive-pair mask*, a matrix of the form

$$Y_{ij} = \begin{cases} 1, & \text{if } p_j \in P^{+}(m_i) \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

Multi-positive molecule-to-pocket loss. Let $S_{ij} = s(p_j, m_i)$ be the similarity score assigned by the model to pocket p_j and molecule m_i. Let τ be the contrastive temperature. For each query molecule m_i in the batch, we define the multi-positive molecule-to-pocket loss as

$$L_i^{m \to p} = -\log \frac{\sum_j Y_{ij} \exp(S_{ij}/\tau)}{\sum_j \exp(S_{ij}/\tau)} \tag{7}$$

This expression differs from the original DrugCLIP loss because the numerator may contain more than one positive pocket for the same query molecule. If only one positive pocket for m_i is present in the batch, Eq. (7) reduces to the usual single-positive contrastive form. If several known positive pockets are present, all of them contribute to the numerator.

The intuition is that, for each query molecule, the softmax distribution is computed over the pockets in the batch, but the probability mass assigned to all known positive pockets is added together. Thus, the loss does not force the model to choose only one correct pocket for a molecule. Instead, it rewards the model whenever the known positive pockets for that molecule receive high scores relative to the other candidate pockets in the batch. The molecule-to-pocket loss for the full batch is obtained by averaging over the molecules:

$$L^{m \to p} = \frac{1}{N} \sum_{i=1}^{N} L_i^{m \to p} \tag{8}$$

Pocket-to-molecule loss. Although the main objective of BioSensIA-DC is target fishing, the contrastive objective can also include the reverse direction. In this case, each pocket is used as a query, and the molecules in the batch are treated as candidates. Using the same similarity matrix S and the same positive-pair mask Y, the pocket-to-molecule loss for pocket p_j is

$$L_j^{p \to m} = -\log \frac{\sum_i Y_{ij} \exp(S_{ij}/\tau)}{\sum_i \exp(S_{ij}/\tau)} \tag{9}$$

This term is analogous to Eq. (7), but it normalizes over molecules rather than over pockets. It rewards the model when the molecules known to be positive for pocket p_j receive high scores relative to the other molecules in the batch.

Let B be the number of pockets in the batch. The average pocket-to-molecule loss is

$$L^{p \to m} = \frac{1}{B} \sum_{j=1}^{B} L_j^{p \to m} \tag{10}$$

Total fine-tuning loss. The total loss for a batch can be written as a weighted combination of the two retrieval directions:

$$L = \lambda_m L^{m \to p} + \lambda_p L^{p \to m} \tag{11}$$

where λ_m and λ_p control the relative importance of the two terms. For a target-fishing-oriented model, it is natural to give more weight to the molecule-to-pocket direction, using $\lambda_m > \lambda_p$, or even to set $\lambda_p = 0$ in an initial experiment. Keeping a nonzero value of λ_p is also a reasonable variant, because it preserves a closer connection to the bidirectional contrastive training used by DrugCLIP.

Fine-tuning variants. Several fine-tuning variants can be derived from the same loss formulation. The most conservative strategy is to freeze both encoders and update only the final projection layers that map molecule and pocket representations into the shared latent space. This reduces the number of trainable parameters and limits the risk of damaging the representations learned during the original DrugCLIP training.

A second variant is to train the projection layers together with the temperature parameter τ. Since τ controls how concentrated the softmax distribution is, adjusting it during fine-tuning may help adapt the ranking behavior to the target-fishing setting.

A less conservative strategy is to unfreeze the upper layers of the encoders, allowing the model to modify part of the molecule and pocket representations while still preserving the lower-level 3D features.

The most flexible strategy is to fine-tune the full model with a smaller learning rate, but this also has the greatest risk of overfitting or catastrophic distortion of the original latent space.

These variants can be combined with different choices of loss direction. One can train with the molecule-to-pocket loss alone, which directly matches the target-fishing task, or with the symmetric loss in Eq. (11), which also includes the pocket-to-molecule term, in which case a grid of values for λ_m and λ_p may be tried.

Comparing these alternatives is part of the diagnostic work needed to determine whether the proposed formulation improves molecule-to-pocket retrieval in practice.

3. RESULTS AND DISCUSSION

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4. CONCLUSIONS

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Acknowledgements

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